

Total No. of Questions :10]

SEAT No. :

P3095

[5461]-132

[Total No. of Pages : 3

**B.E. (Mechanical Engineering)
CAD/CAM AND AUTOMATION
(2012 Course) (Semester-I) (402042)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.No.6, Q.No.7 or Q.No.8, Q.No.9 or Q.No.10.
- 2) Figures to the right indicate full marks.
- 3) Use of Electronic pocket calculator is allowed.
- 4) Assume suitable data, if necessary.

Q1) a) A rectangle ABCD with vertices A(3,2), B(5,2), C(5,7) and D(3,7) is rotated through 90° clockwise about vertex A. Find the coordinates of new vertices. **[6]**

b) Compare with neat sketch, Bezier, B spline and Cubic curves. **[6]**

OR

Q2) a) Compare Analytical Curves with Synthetic curves. **[4]**

b) Three tension springs with stiffness 21 N/mm are arranged once in series and then in parallel. Compare the deflection of free end if tensile force applied at free end is 2.1 kN. **[8]**

Q3) a) A line with vertices (10, 10) and (10, 50) is mirrored about y axis. Find the new vertices. **[4]**

b) Explain relation between Intrinsic and Global coordinate system in FEM. **[4]**

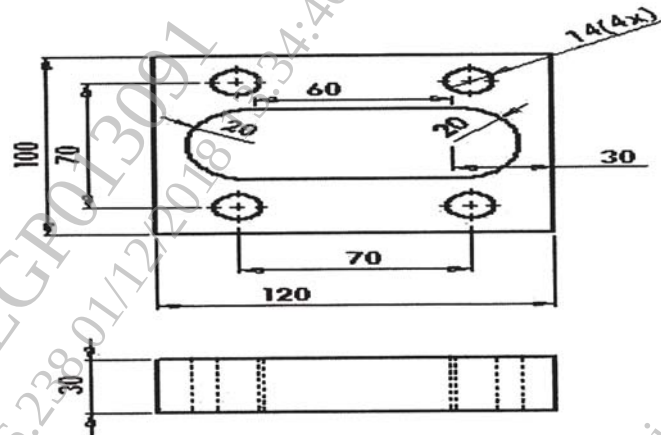
OR

Q4) a) Write short note on CSG. **[4]**

b) Explain the need of defining a shape function in FEM. **[4]**

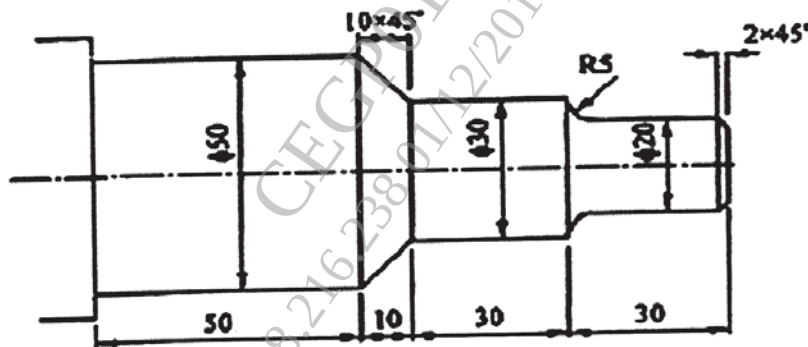
P.T.O.

- Q5) a)** Explain the steps to be followed in developing a CNC part program. [8]
- b) Write a CNC program using G and M codes for contouring a component shown in following figure. Assume suitable data for speed, feed and depth of cut. [10]



OR

- Q6) a)** Write a CNC program using G and M codes to turn a component shown below. Assume suitable data for speed, feed and depth of cut. Use only canned rough turn and finish cycles. [10]



- b) Write G codes for: [8]

X-Y plane selection
Feed Per Minute
Feed Per Revolution
Cancel Canned Cycle

Write M codes for :

Coolant on
End of program
Tool change
Spindle stop

- Q7) a)** Explain Fused Deposition Modeling (FDM) in detail with neat sketch. State its advantages and applications. **[8]**
- b)** Explain 3D printing process in detail with neat sketch. State its advantages and applications. **[8]**

OR

- Q8) a)** Explain Stereo Lithography technique in detail with neat sketch. State its advantages. **[8]**
- b)** Explain Selective Laser Sintering (SLS) in detail with neat sketch. State its limitations and applications. **[8]**

- Q9) a)** Differentiate between Point to point & continuous path robotic systems. **[8]**
- b)** Write short note on FMS. **[8]**

OR

- Q10) a)** Write short note on Group Technology. **[8]**
- b)** Explain Robot Anatomy with the help of supporting sketches. **[8]**

